

- [Charts: visualizing data](#)
- [Getting started with BigQuery](#)

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the [full course website](#) for more.

- [Intro to Pandas DataFrame](#)
- [Intro to RAPIDS cuDF to accelerate pandas](#)
- [Getting Started with cuML's accelerator mode](#)

Using Accelerated Hardware

- [Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API](#)
- [Train a Vision Transformer \(ViT\) for image classification with JAX](#)
- [Text classification with a transformer language model using JAX](#)

Featured examples

- [Train a miniGPT language model with JAX AI Stack](#)
- [LoRA/QLoRA finetuning for LLM using Tux](#)
- [Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA](#)
- [Loading Hugging Face Transformers Checkpoints](#)
- [8-bit Integer Quantization in Keras](#)
- [Float8 training and inference with a simple Transformer model](#)
- [Pretraining a Transformer from scratch with KerasHub](#)
- [Simple MNIST convnet](#)
- [Image classification from scratch using Keras 3](#)
- [Image Classification with KerasHub](#)

```
import numpy as np
import matplotlib.pyplot as plt

sales_data = np.array([
    [250, 300, 280],
    [450, 500, 475],
    [150, 180, 170]
])

products = ["Product A", "Product B", "Product C"]

average_prices = np.mean(sales_data, axis=1)

print("Average price of each product:")
for product, avg in zip(products, average_prices):
    print(f"{product}: {avg:.2f}")

overall_average = np.mean(sales_data)

print(f"\nAverage price of all products sold: {overall_average:.2f}")

plt.figure(figsize=(8,5))

bars = plt.bar(products, average_prices)

highest = np.argmax(average_prices)
bars[highest].set_color("orange")

for bar in bars:
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 5,
        f"{bar.get_height():.2f}",
        ha="center"
    )

plt.title("Average Price of Each Product")
```

```
plt.xlabel('Products')  
plt.ylabel("Average Price")  
plt.ylim(0, max(average_prices) + 100)  
plt.grid(axis="y", linestyle="--", alpha=0.7)  
  
plt.show()
```

Average price of each product:

Product A: 276.67

Product B: 475.00

Product C: 166.67

Average price of all products sold: 306.11

